

Russell Rosa

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EDUCATION

Purdue University, College of Science

West Lafayette, IN

Bachelor of Science, Data Science and Computer Science GPA: 3.3

May 2026

PROFESSIONAL EXPERIENCE

Peterson Technology Partners

Park Ridge, IL

Data Science Intern

Jul 2025 – Aug 2025

- Engineered a scalable data collection pipeline using Python, leveraging Playwright and search APIs to extract and process 1000+ resumes within a week, while ensuring compliance with regulatory requirements in Insurance Operations environments.
- Developed and implemented a machine learning-based document classification system using Sentence-BERT embeddings for data analysis, easyOCR text extraction for data preprocessing, and Random Forest models for statistical analysis, achieving approximately 90% accuracy in data modeling and descriptive statistics.
- Designed an end-to-end ETL workflow in Jupyter Notebook with SQLite caching to support Data Warehousing and Business Intelligence initiatives, automated feature extraction for enhanced Data Modeling, and structured analytics dashboards for reporting Key Performance Indicators (KPIs) on candidate sourcing insights.

The Data Mine Corporate Partners

West Lafayette, IN

Undergraduate Data Science Researcher

Aug 2022 – Apr 2023

- Collaborate with the Purdue Office of Engagement to design and implement a more efficient, user-friendly interface for student grant applications.
- Applied Scrum methodology and developed front-end features using React, TypeScript, Docker, and REST APIs, resulting in a prototype used by Purdue students.

CLASS AND PERSONAL PROJECTS

AI-Driven Dynamic Pricing for Autonomous Delivery

2026

- Built predictive models using structured data to simulate real-world decision systems and estimate demand through statistical analysis, applying machine learning techniques to optimize pricing strategies in compliance with regulatory requirements for insurance operations.
- Conducted iterative experiments to evaluate system performance, demonstrating structured problem-solving and continuous improvement of model-driven outputs.
- Implemented and compared static and adaptive pricing strategies using data analysis techniques to evaluate revenue growth, customer wait time optimization, and congestion management tradeoffs in a simulated environment, resulting in informed business decisions for BI Analysts and Data Scientists.

AI Story-to Video Automation Pipeline

2025

- Developed a fully automated AI-powered application that integrates data analysis and business intelligence capabilities by processing raw text through Large Language Model-driven statistical analysis, generating key performance indicators (KPIs) and aligned subtitles for enhanced data visualization.
- Designed and implemented modular pipeline components using Data Modeling techniques to manage sequential transformations in a scalable system design, integrating with APIs for orchestration and ensuring seamless data flow through REST-based workflows.

Object Detection, OCR, and Grade Prediction for ER Diagrams

2024

- Developed and implemented a machine learning-based object detection module in Power BI to accurately identify and classify components within Entity Relation Diagrams (ERDs), utilizing statistical analysis techniques for precise classification and data visualization to inform business intelligence.
- Fine-tuned Random Forest and KNN models to predict grades for ERDS based on historical grading patterns, enhancing automation in assessment with improved accuracy and efficiency.

LEADERSHIP AND INVOLVEMENT

Purdue Filipino Association

Aug 2022 – Present

President

- Developed and executed comprehensive event planning strategies for 200+ students, successfully leading a team of 20+ members to deliver cultural, social, and philanthropic events that fostered engagement and community building within the student body.
- Presented data-driven plans to board members and coordinated large-scale events, showcasing strong communication, organization, and client-facing skills while ensuring regulatory compliance and meeting key performance indicators (KPIs) for insurance operations.

Asian Student Union Board

Aug 2023 – May 2025

Vice President Internal

- Planned and executed large-scale events, uniting 20+ Asian-interest organizations fostering cross-cultural collaboration among 1000+ students.
- Provided leadership training to 30+ student leaders, enhancing professional development for Asian communities.

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, C#, JavaScript, TypeScript, Bash

ML: Pandas, NumPy, Scikit-learn, Random Forest, KNN, EasyOCR

Data: Data Analysis, Data Validation, Data Cleaning, Trend Analysis, Statistical Analysis, Data Visualization
Tools & Infrastructure: Git, Docker, Playwright, FFmpeg, VS Code, Terminal, Jupyter Notebook, Databricks